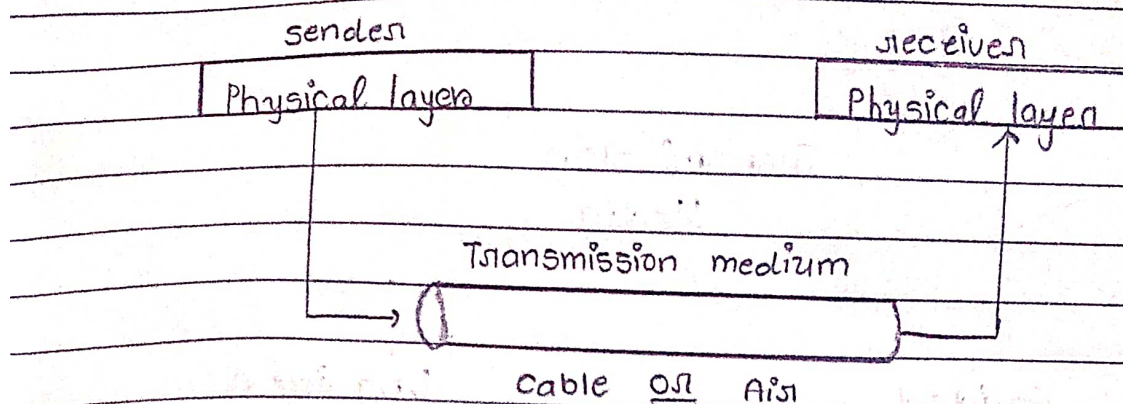


UNIT-3 Transmission Media & Network devices

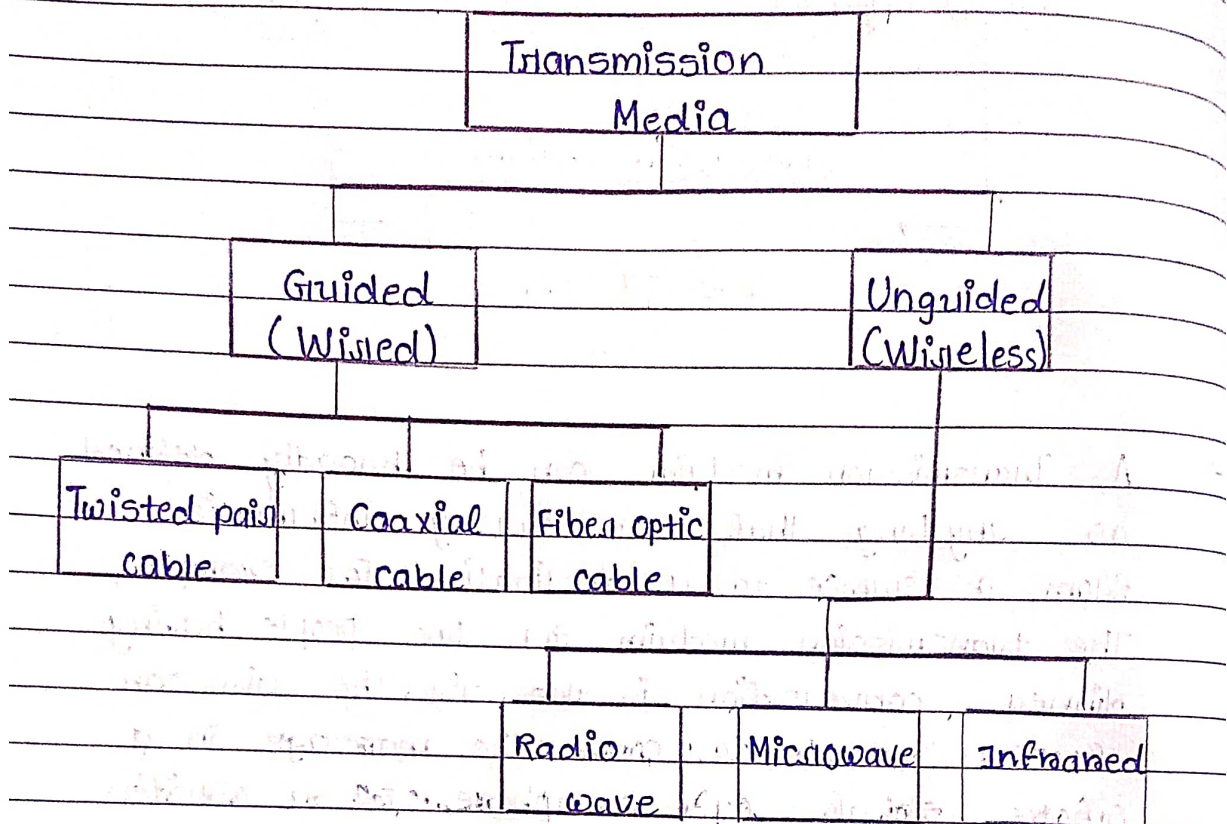
1* Transmission Medium :



→ A Transmission medium can be broadly defined as anything that can carry information from a source to a destination. For example, The transmission medium for two people having dinner conversation is the air. The air can also be used to convey the message in a smoke signal or semaphore. For a written message, the transmission medium might be mail carrier, a truck or a airplane.

→ In data communications the transmission medium is usually a free space, metallic cable or Fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form.

* 2 Classes of transmission media



9.1

* Guided Media :

→ Guided Media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium.

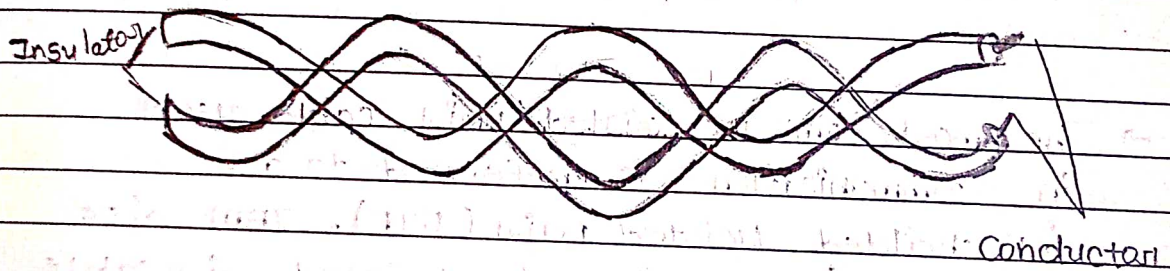
→ Twisted pair and coaxial cable use metallic conductors that accept and transport signal in the form of electric current. Optical fiber is a glass or

plastic cable that accepts and transport signal in the form of light.

2.2 * Unguided Media :

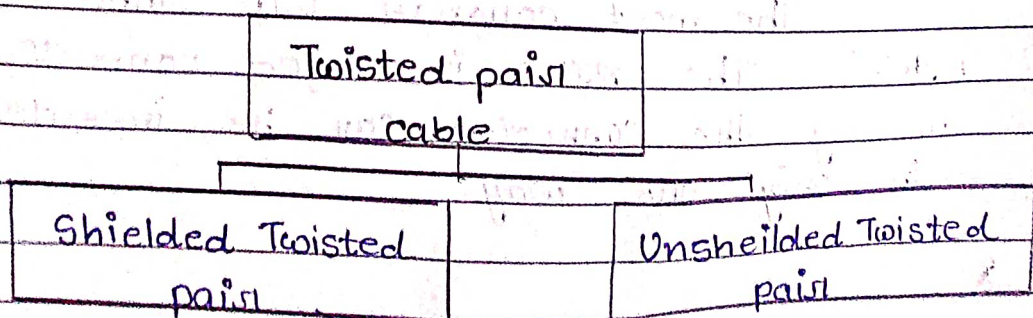
→ Unguided media or wireless communication, transport electromagnetic waves without using physical conductor, Instead signals are broadcast through air and thus are available to anyone who has a device capable of receiving them.

* 3 Twisted - pair cable :



→ It consist of two conductors, each with its own plastic Insulator, twisted together. One of the wire is used to carry signals to receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

* 3.1 Types of Twisted pair cable:

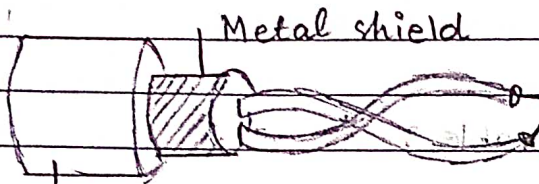


Unshielded Twisted pair :



Plastic cover

Shielded Twisted pair :

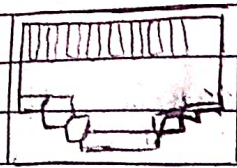


Plastic cover

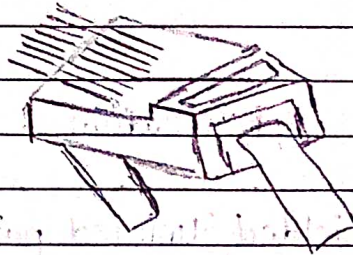
→ The most common twisted-pair cable used in communication is referred to as "Unshielded twisted pair (UTP)". IBM also produced a version of twisted pair cable for its use, called shielded twisted pair (STP). STP cable has a metal foil or a braided mesh covering that encases each pair of insulator conductors.

*3.2 Connectors :

The most common UTP connector is "RJ45". The RJ45 is keyed connector meaning the connector can be inserted in only one way.



RJ-45 Female

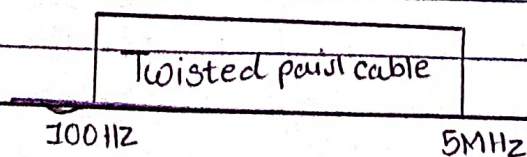


RJ-45 Male

*3.3 Applications :

- (i) Twisted pair cable is used in telephone lines to provide voice and data channels.
- (ii) The local-loop = The line that connects subscribers to the central telephone office commonly consist of unshielded twisted pair cable.
- (iii) The DSL lines that are used by the telephone companies to provide high data rate connections also use the high-bandwidth capability of unshielded - pair cable.

3.4* Frequency Range :

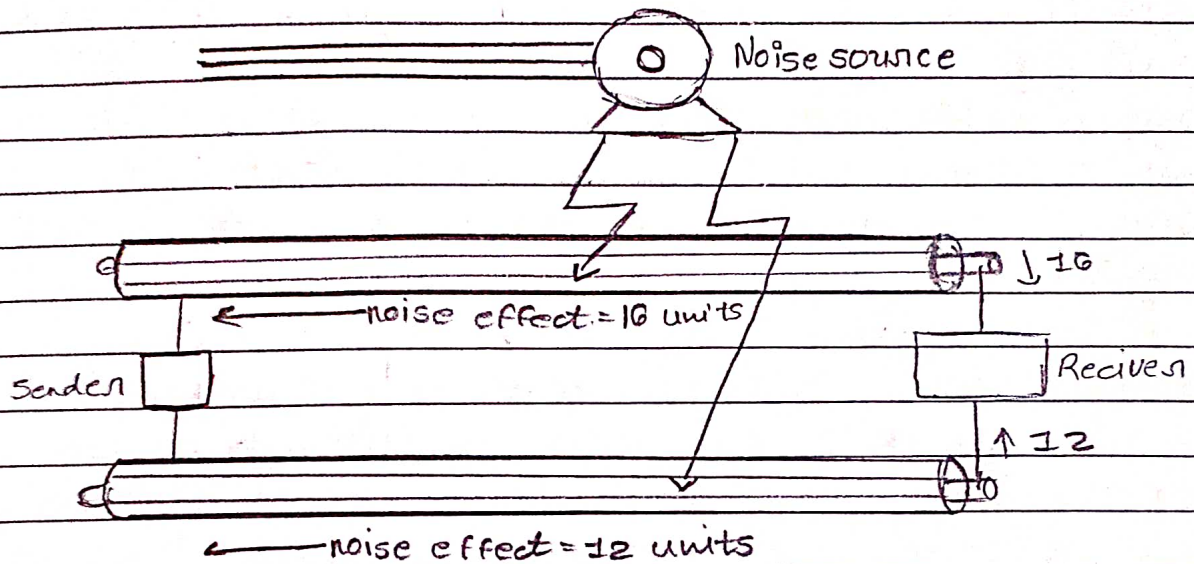


3.5 * Categories of twisted pair cable :

Category	Specification	Data Rate	USE
1	Unshielded twisted pair used in telephone	<0.1	Telephone
2	Unshielded twisted pair is originally used in T lines	2	T-1 lines
3	Improved CAT2 used in LAN's	10	LAN's
4	Improved CAT3 used in Token Ring network.	20	LAN's
5	Cable wire is normally AWG 24 with a jacket and outside sheath	100	LAN's
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference.	125	LAN's

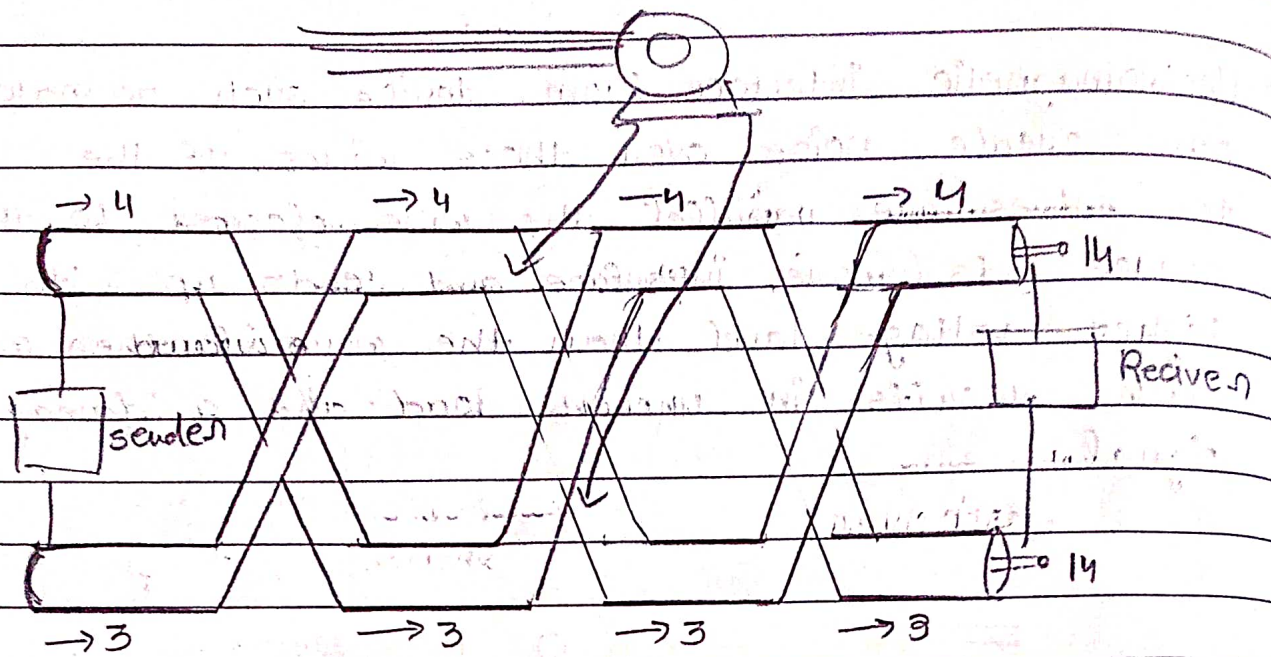
* Why twisted pair cable? :

→ Electromagnetic interference from device such as motor can create noise over those wires. If the two wires are parallel, the wire closest to the source gets more interference and ends up with higher voltage level than the wire further away, which results in uneven load and a damaged signal.



→ Now, If the both wires are twisted then the cumulative effect of the interference is equal on both wires.

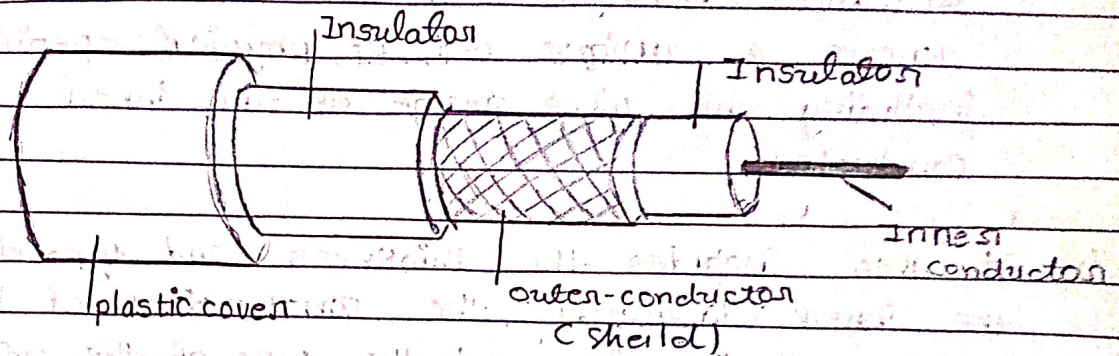
→ The total effect of the noise at receiver is therefore 0. Twisting does not always eliminate the impact of noise, but it does significantly reduce it.



$$\text{SO, } 14 - 14 = 0$$

4

* Coaxial Cable :



(*) Coaxial cable carries signals of higher frequency ranges than those in twisted-pair cable, in part because the two media are constructed quite differently.

(*) Instead of having two wires, coax has a central core conductor of solid or stranded wire enclosed in an insulating sheath which is in turn, encased in an outer conductor of metal foil, or combination of two.

(*) The outer metallic wrapping serves both as a shield against noise and as the second conductor which completes the circuit.

(*) This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by plastic cover as shown in upper fig.

4.1

↳ Standards of coaxial cable :

- (*) Coaxial cable are categorized by their Radio Government ratings. Each RG number denotes a unique set of physical specification including the wire gauge of the inner conductor.
- (*) It also includes the thickness and type of the inner insulator, the construction of the shield and the size and the type of the outer casing.

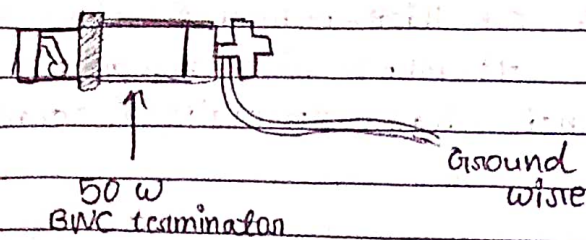
category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

4.2

* Connectors :

- (*) To connect coaxial cable to device, we need connectors. The most common type of connector used is " Bayonet Neil - Concelman (BNC) connector.





(i) BNC connector is used to connect the end of the cable to a device, such as a TV set.

(ii) It is also used in Ethernet networks to branch out to a connection to a computer or other device.

4.3

* Applications :

(i) Coaxial cable was widely used in analog telephone networks where a single coaxial network could carry 10,000 voice signals.

(ii) Coaxial cable in telephone networks has largely been replaced today with fiber-optic cable.

(iii) Another common application of coaxial cable is in traditional Ethernet LANs. Because of its high bandwidth, and the consequently high data rate it was chosen for digital transmission in early Ethernet LANs.

(iv) The 10base-2 or thin ethernet uses RG-58 coaxial cable with BNC connector to transmit data at 10 Mbps with a range of 185 m.

(*) The 10Base5 or thick ethernet uses RG11 to transmit 10 mbps with range of 5000m. Thick ethernet has specialized connectors.

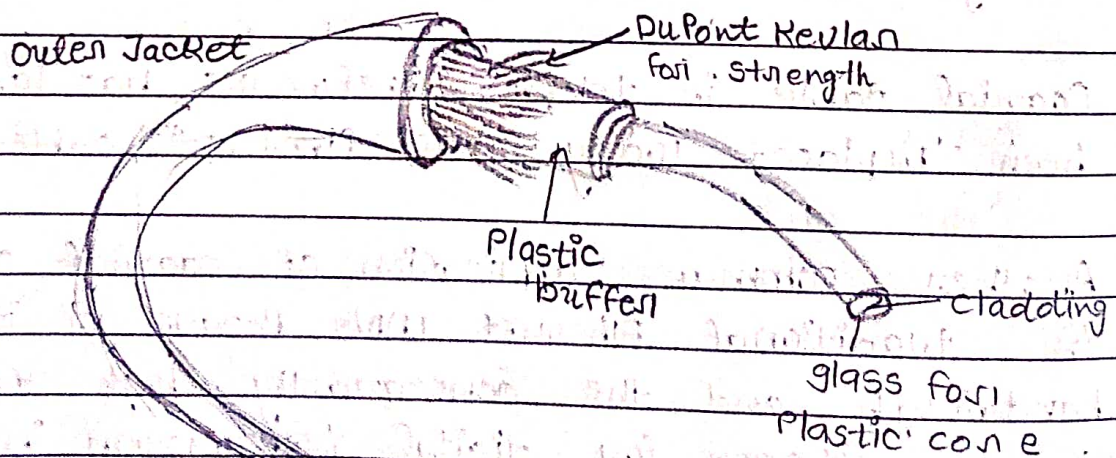
4.4 Frequency range :

coaxial cable

100 KHZ

500 KHZ

* Optical Fiber cable :



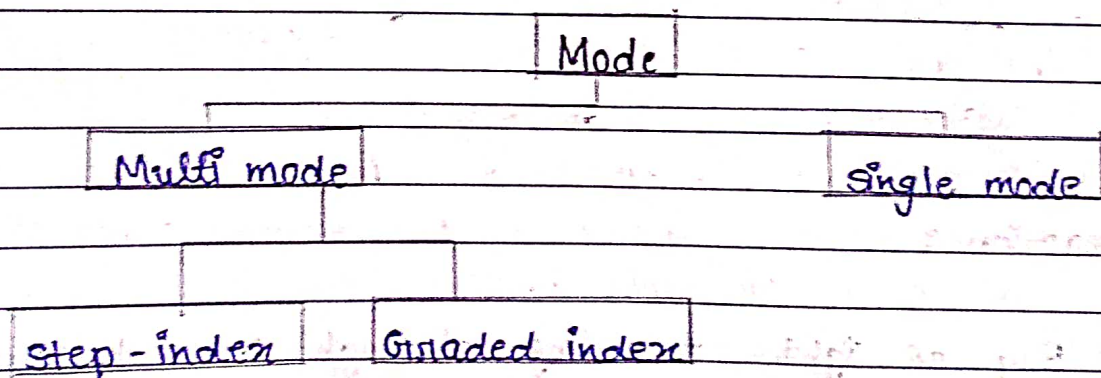
(2) Reflection :

- (1) When the angle of incidence becomes greater than the critical angle, a new phenomenon occurs called Reflection.

(3) Critical Angle :

- (1) The change in the incident angle results in a refracted angle of 90 degrees, with the refracted beam now lying along the horizontal.
- (2) The incident angle at this point is known as the critical angle.

* Propagation Modes :

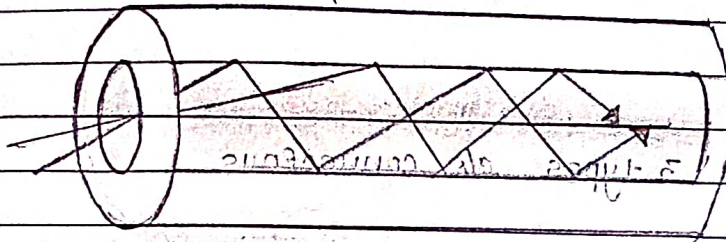


(1) Multimode :

→ It is named multimode because multiple beams from a light source move through the core in different paths.

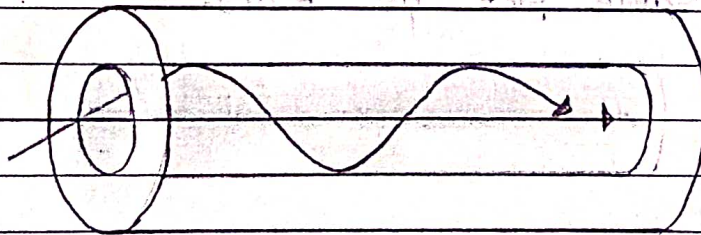
(1) Step index fiber :

- (i) A beam of light moves through this constant density in a straight line until it reaches the interface of core and the cladding.



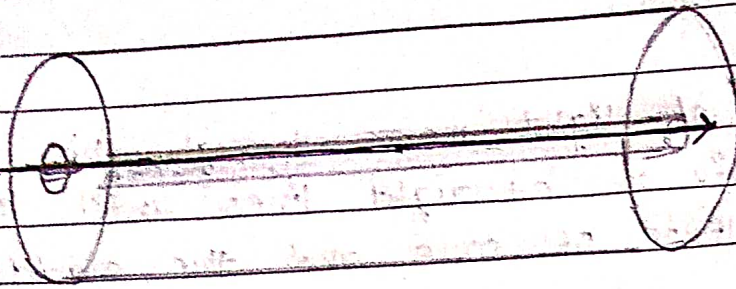
(2) Graded index Fiber:

- (i) Graded index fiber is one with varying densities. Density is highest at the center of the core and decreases gradually to its lowest at the edge.



(3) Single Mode :

- (i) It uses step index fiber and highly focused source of light that limits beam to small range of angles, all close to horizontal.



* Connectors:

(1) There are 3 types of connectors.

(1) Subscriber channel (SC)

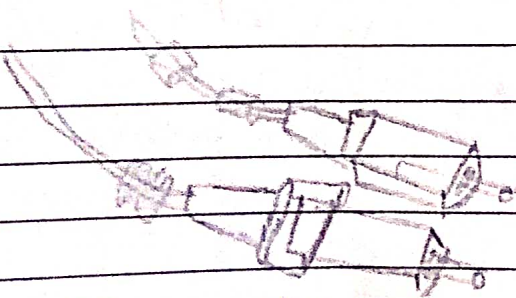
(i) Used for cable TV and uses PUSH/PULL locking system.

(2) Straight tip (ST)

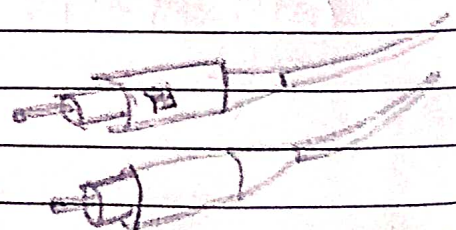
(i) Used to connect networking device and uses bayonet locking system.

(3) MT-RJ

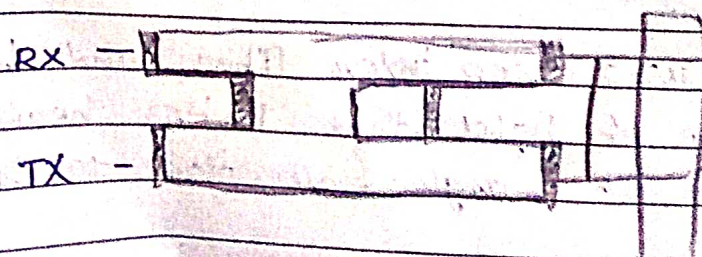
(i) It is same size as RJ45.



SC connector



ST Connector



MT-RJ connector

* Applications :

- (*) It is often found in backbone network because its wide bandwidth is cost-effective.
- (*) Some cable TV companies use a combination of optical fiber and coaxial cable thus creating a hybrid network.
- (*) This cost-effective configuration since the narrow bandwidth requirement at the user end does not justify the use of optical fiber.

* Advantages :

- (*) Higher bandwidth
- (*) Less signal attenuation
- (*) Immunity to electromagnetic interference
- (*) Resistance to corrosive materials.
- (*) Light weight
- (*) Greater immunity to tapping.
- (*) Installation and maintenance.
- (*) Unidirectional light propagation.
- (*) Cost

* Unguided Media :

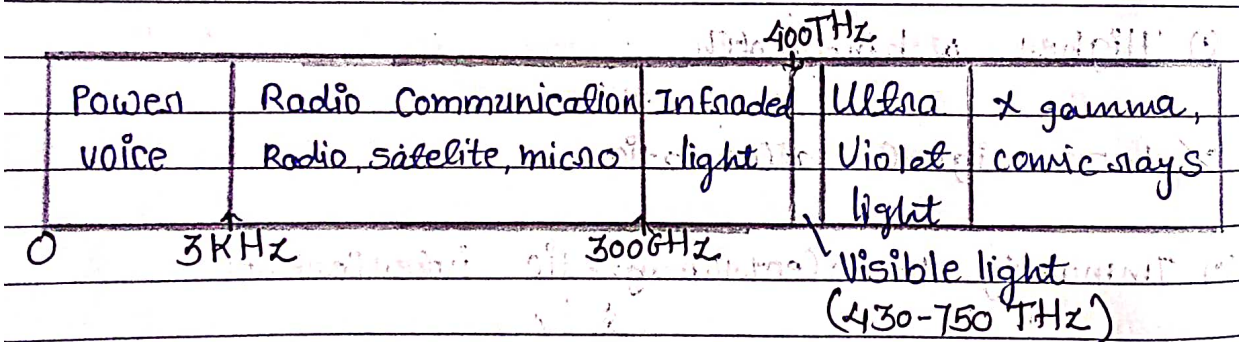
(1) This transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as "wireless communication".

(2) Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

(1) Radio wave

(2) Micro wave

(3) Infrared wave



* Propagation methods :

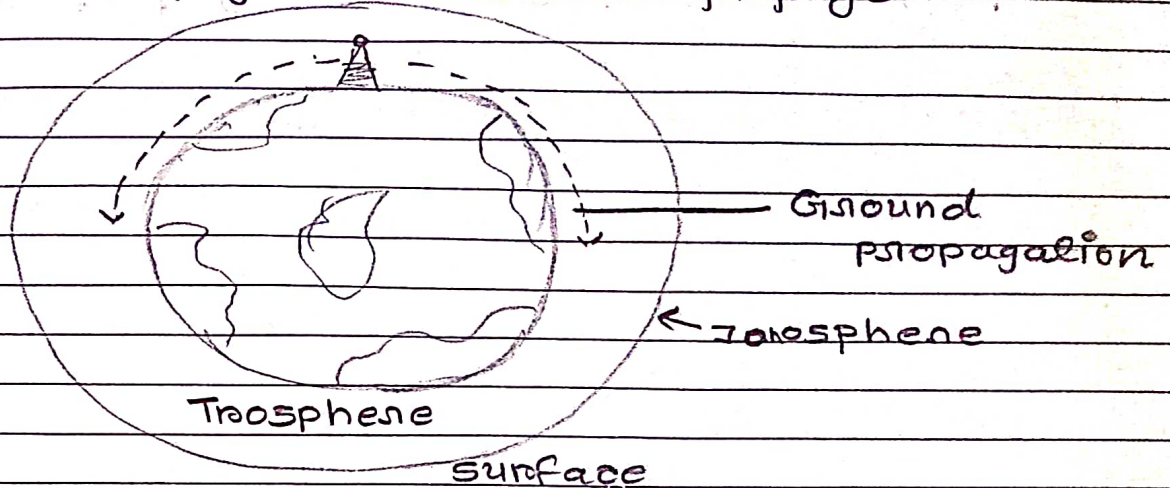
(1) Surface propagation / ground

(2) Tropospheric propagation / sky

(3) Line-of-sight propagation

(4) Space propagation

(1) Surface propagation : / Ground propagation:



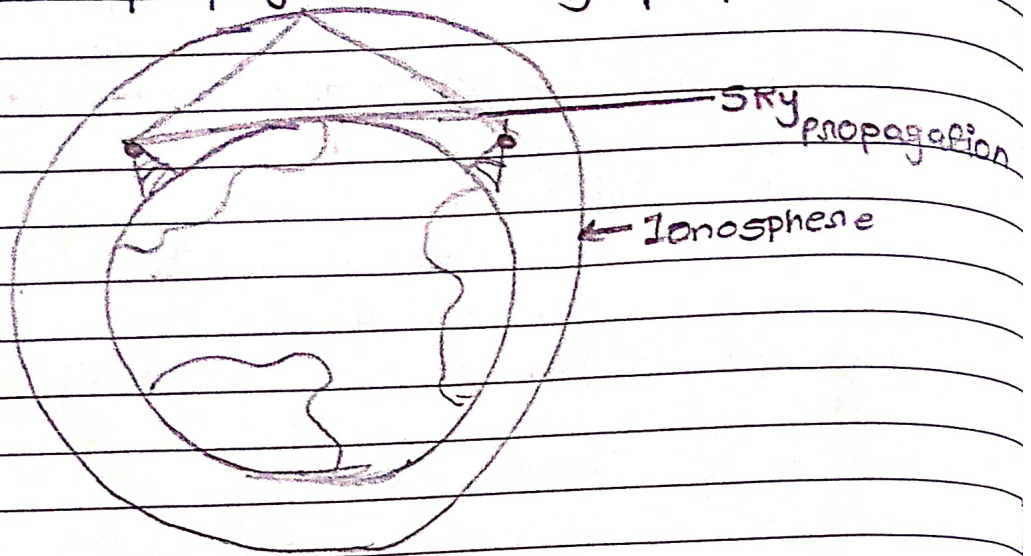
- (i) In this, Radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- (ii) At the lowest frequencies, signals emanate in all direction from the transmitting antenna and follow the curvature of the planet.
- (iii) Distance dependent on the amount of power in the signal: greater the power greater the distance.

→ Application

→ Long range radio navigation

→ Radio beacons and navigational locators.

(2) Tropospheric propagation : / sky propagation



(*) This works in 2 ways: Either a signal can be directed in a straight line from antenna to antenna, or it broadcast out an angle into the upper layers of the troposphere where it is reflected back down to the earth's surface.

(*) The first method is used / requires the placement of receiver and transmitter be within line of straight distance

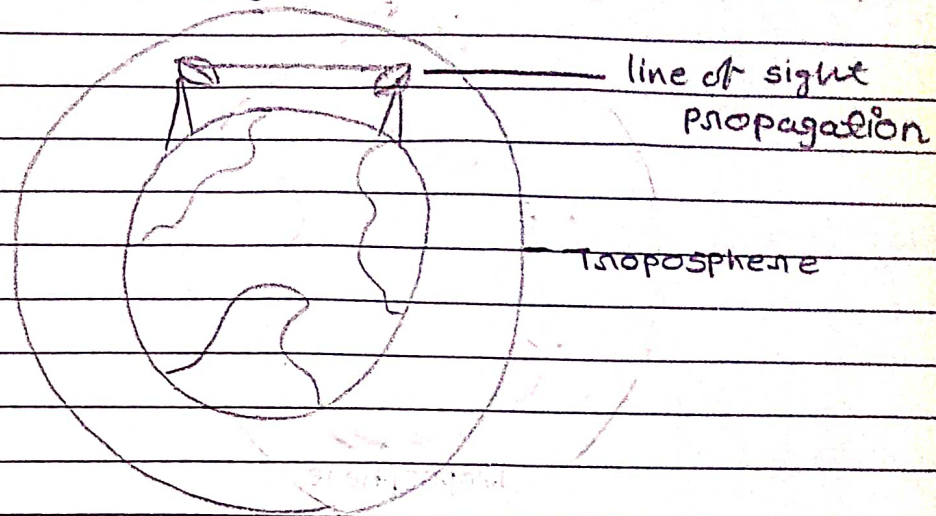
(*) The second method allows greater distance to be covered.

* Applications:

(*) AM Radio

(*) Citizen band, ship / aircraft

* Line-of-sight propagation :

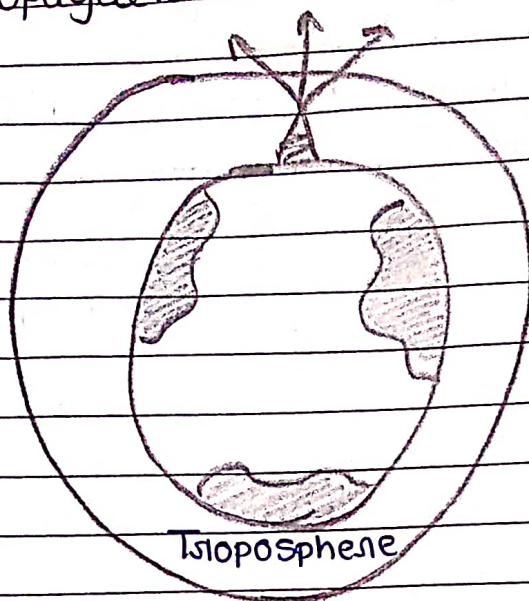


- (i) In this, very high frequency signals are transmitted in straight line directly from antenna to antenna.
- (ii) Line of sight propagation is tricky because radio transmission can not be completely focused. waves emanate upward and downwards as well as forward and can reflect off the surface of the earth.
- (iii) Reflected waves that arrive at the receiving antenna later than the direct portion of the transmission can corrupt the received signals.

→ Application :

- (i) VHF TV, cellular phones, satellite
- (ii) Radar, satellite
- (iii) paging, FM radio.

(4) Space propagation :



(i) It utilizes satellite relays in place of atmospheric refraction. A broadcast signal is by an orbiting satellite, which rebroadcasts the signal to the intended receiver by an orbiting satellite earth.

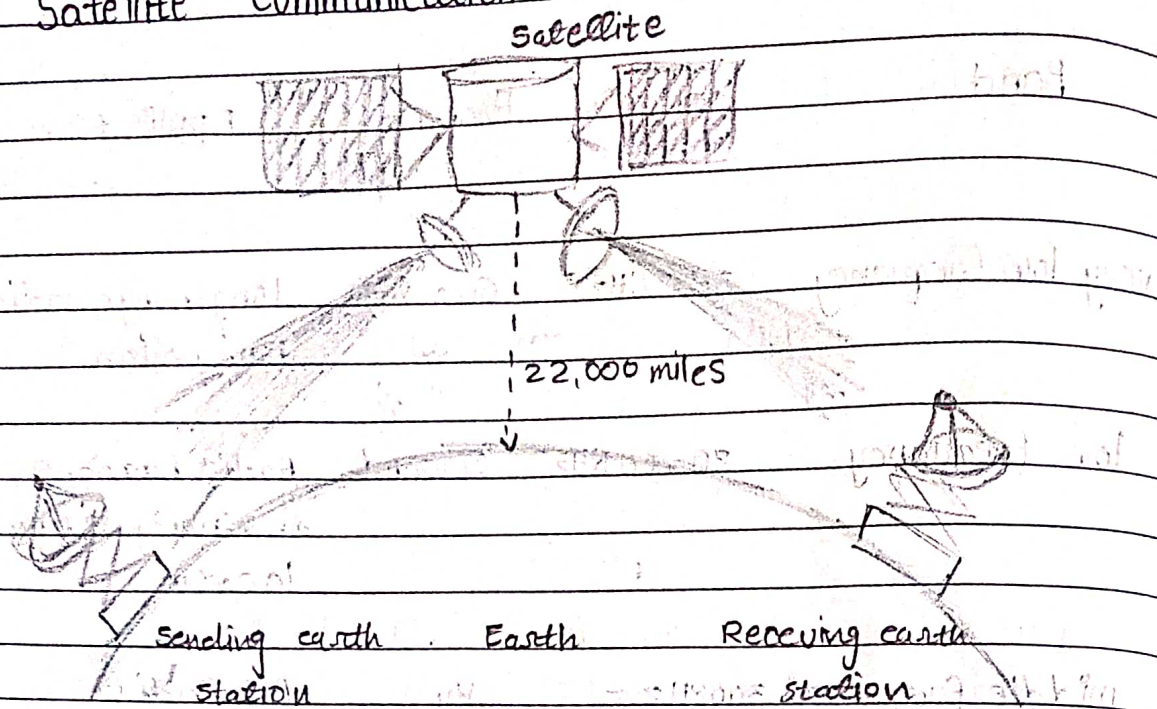
(ii) Which is the distance of the satellite from the earth makes it the equivalent of a super high gain antenna and dramatically increases the distance coverable by a signal.

• Application :

- Distv Dish TV
- Satellite Satellite

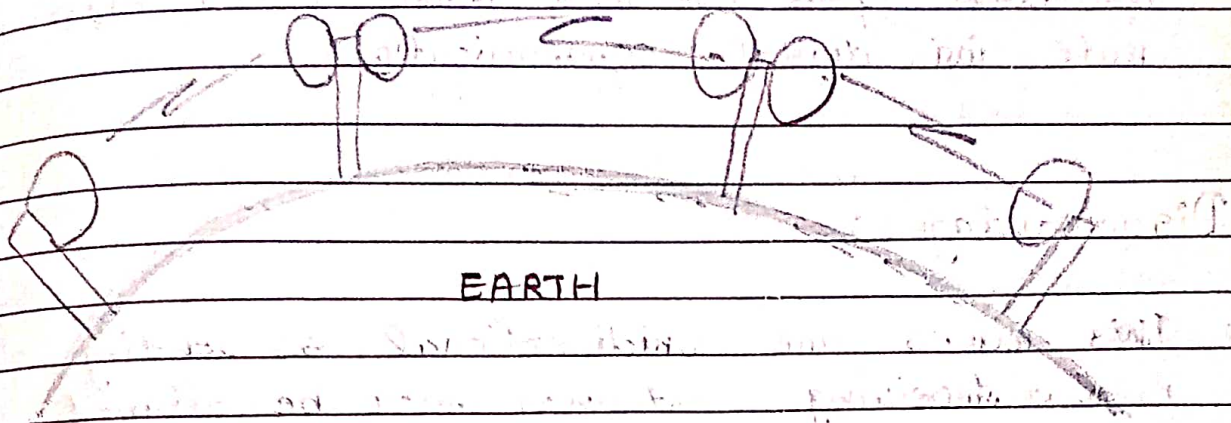
Band	Range	Propagation	Application
very low frequency	3-30 KHz	Ground	long-range radio navigation
low frequency	30-300 KHz	Ground	Radio beacons and navigational locations.
middle frequency	300 KHz - 3 MHz	Sky	AM Radio
high frequency	3-30 MHz	Sky	citizen band, ship / aircraft
very high frequency	30-300 MHz	sky and line of sight	VHF, FM radio
ultra high frequency	300 MHz - 3 GHz	line of sight	UHF-TV, cellular phone, paging
super high frequency	3-30 GHz	line of sight	satellite
extremely high frequency	30-300 GHz	line of sight	Radar, satellite

* Satellite Communication :



- Satellite transmission is much like line of sight microwave transmission in which one of the stations is a satellite orbiting the earth.
- In satellite transmission signals must travel in straight lines.
- Satellite microwave can provide transmission capability to and from any location on earth no matter how remote.
- This advantage makes high quality communication available to underdeveloped parts of the world without requiring a huge investment in ground based infrastructure.
- Satellite themselves are extremely expensive, of course but leasing time on frequency or one can be relatively cheap.

* Terrestrial Microwave: (1GHz To 300GHz)



- Microwaves do not follow the curvature of the earth and therefore requires line of sight transmission and reception equipment.
- Microwave signals propagate in one direction at a time, which means that two frequencies are necessary for two way communication such as telephone conversation.
- One frequency is reserved for microwave transmission in one direction and the other for transmission in other.
- Each frequency requires its own transmitter and receiver.
- Today, both pieces of equipment usually are combined in a single piece of equipment called transceiver.

◦ Advantages :

→ There is no interference of another pair of aligned antennas.

→ As band is wider than radio waves, high rate for digital communications.

◦ Disadvantages :

→ These waves are unidirectional, so sending and receiving antennas must be aligned.

→ Repeaters are required for long distances.

◦ Applications :

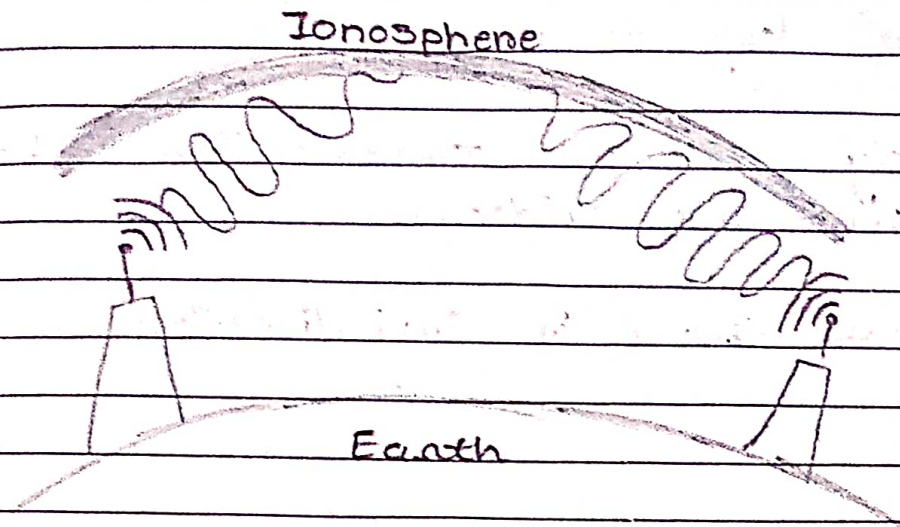
→ Used for unicasting.

→ cellular phones

→ Satellite networks

→ Wireless LANs

* Radio wave transmission :



- Radio waves are electromagnetic ^{waves} occurring on radio frequency portion of the electromagnetic spectrum.
- These electromagnetic waves are ranging in frequency between 3 kHz and 1 GHz.
- A common use is to transport information through the atmosphere or outer space without wires.
- Radio waves are omni directional means they can travel in all direction from the source. So transmitter and receiver do not need to be aligned physically.
- Radio waves are of "sky propagation".
- They are easy to generate, can travel long distance and can penetrate buildings.

• Advantages :

→ Travel long distance

→ widely used for communication in both indoor and outdoors.

→ It can travel in Omni direction

• Disadvantages :

• Disadvantages :

→ Due to the narrow side bands, there is low data rate for digital communication.

→ They can penetrate walls so we can not isolate whether the communication is just inside or outside the building.

• Applications :

→ AM and FM radio

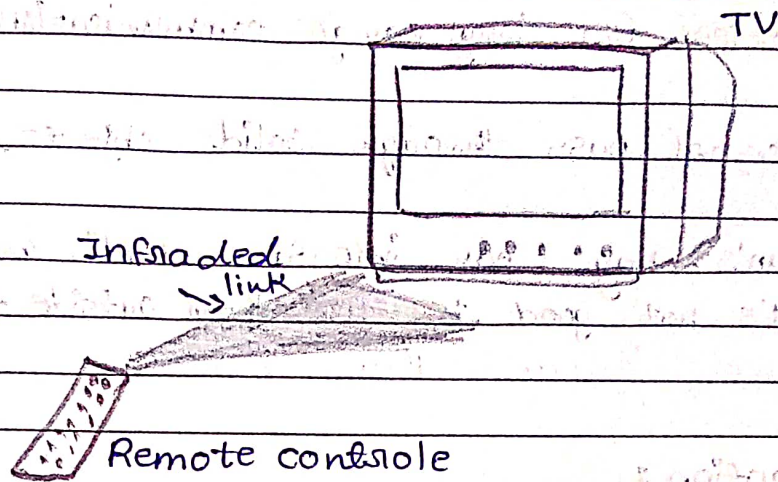
→ GPS service

→ cordless phone

→ police radio

→ Television broadcast.

* Infrared waves transmission :



(i) These electromagnetic waves are ranging in frequency between 300 GHz and 400 THz.

(ii) They are used for short range communication.

(iii) Example : remote control used in televisions, VCRs, stereos.

(iv) They are line of sight propagation.

* Advantages :

→ They are relatively directional, cheap and easy to built.

→ Having high frequency so because of that they are not able to penetrate the walls.

* Disadvantages :

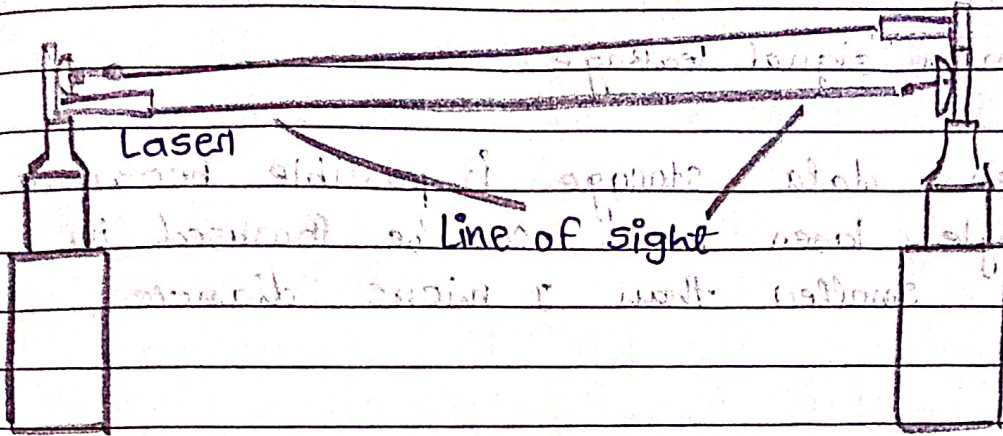
- Useless for long range communication.
- Do not pass through solid objects.
- Sun's ray can interfere radio waves,
It's not good to use them outside the building

* Application :

- Remote control of TV, VCR
- Thermal efficiency analysis, remote temperature sensing.
- military application: target acquisition, surveillance tracking.

* Laser transmission:

Photo detector



(1) Full form: Light Amplification Stimulated Emission or Radiation.

(2) They are unidirectional, Therefore this type of system use "line of sight" propagation.

(3) It can Transmit signal to long term.

(4) They can not penetrate obstacles such as walls, rain, thick fog. laser beam is distorted by wind, atmosphere temperature.

(5) A photo detector and laser is set on both sender and receiver side.

* Advantages:

(1) It is safe for data transmission as it is 3mm wide laser which is difficult to tap.

(*) High bandwidth at low cost.

(*) High speed.

(*) minimum signal leakage.

(*) Higher data storage is possible because single laser beam can be focused in area smaller than 1 micro diameter.

* Disadvantages :

(*) Maintenance is costly.

(*) harmful to human beings and often burns them during contacts.

* Application :

(*) It has been utilized for mass communication including telephone conversation and even television channels.

(*) Used in digital data transmission.

(*) Effectively used for shorted distance communication.